

## 10.9 Solving Problems Involving Tangents

### Lesson Introduction

|                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                         |                                                                                                     |                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
|  Benchmark           | MA.912.GR.6.2 Solve mathematical and real-world problems involving the measures of arcs and related angles.                                                                                                                                                                                                                                                                                                                                |                                                                                                                         |  Learning Target | <ul style="list-style-type: none"><li>I can solve problems related to tangents and secants and the measures of their intercepted arcs.</li></ul>    |
|  Benchmark Coherence | Building On                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                         | Working Toward                                                                                      |                                                                                                                                                     |
|                                                                                                       | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                        | N/A                                                                                                                     |                                                                                                     |                                                                                                                                                     |
|  Essential Questions | <ul style="list-style-type: none"><li>What is the relationship between a line tangent to a circle and the radius of that circle?</li><li>What is the relationship between the angles created by two secants, two tangents, or one secant and one tangent intersecting at an external point on the circle?</li></ul>                                                                                                                        |                                                                                                                         |                                                                                                     |                                                                                                                                                     |
|  Lesson Narrative    | Throughout this lesson, students are introduced to tangent lines and the angles formed by intersecting tangents and secants. Students will begin by exploring tangent lines and the relationships with the circle the line runs tangent to. They will then collaborate to discover angles formed by intersecting tangents and secants. Students will use this knowledge to solve mathematical problems for missing arc and angle measures. |                                                                                                                         |                                                                                                     |                                                                                                                                                     |
|  Component Summary | 10.9.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                                                                                                   | Decoding a number puzzle ( <i>SE p. 164</i> )                                                                           | 10.9.3 Collaboration (10-15 min.)                                                                   | Discovering the relationship between angles formed by intersecting tangents or tangents and secants and their intercepted arcs ( <i>SE p. 166</i> ) |
|                                                                                                       | 10.9.2 Exploration (5-10 min.)                                                                                                                                                                                                                                                                                                                                                                                                             | Observing relationships of lines drawn from the center of a circle to a line tangent to the circle ( <i>SE p. 165</i> ) | 10.9.4 Your Turn! (10-15 min.)                                                                      | Solving problems involving the intersection of tangents or a tangent and secant ( <i>SE p. 168</i> )                                                |
|                                                                                                       | 10.9.5 Wrap-Up (~5 min.)                                                                                                                                                                                                                                                                                                                                                                                                                   | Reflecting on what is not known yet and what is understood ( <i>SE p. 169</i> )                                         |                                                                                                     |                                                                                                                                                     |
|  Resources         | Glossary                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                         | Materials                                                                                           | Additional Preparation                                                                                                                              |
|                                                                                                       | <u>Key Terms:</u> <ul style="list-style-type: none"><li>Tangent</li><li>External Angle to a Circle</li></ul> <u>Additional Glossary:</u> <ul style="list-style-type: none"><li>N/A</li></ul>                                                                                                                                                                                                                                               |                                                                                                                         | <ul style="list-style-type: none"><li>(Optional) GeoGebra Activities</li></ul>                      | <ul style="list-style-type: none"><li>N/A</li></ul>                                                                                                 |

|                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher<br>Tips | <b>Common Misconceptions</b> <ul style="list-style-type: none"><li>Students may confuse a secant line with a tangent line and vice-versa.</li><li>Students may think the angle relationships are the same as one of the other angle relationships in circles.</li><li>Students may subtract the wrong intercepted arcs.</li></ul>                                                                                                                                                                                                                                                                                                               | <b>Things to Consider</b> <ul style="list-style-type: none"><li>Be cautious not to give away anything in 10.9.2 Exploration, as it is designed to be a discovery activity for students.</li></ul>                                                                                                                                                                            |
|                                                                                                      | <b>Literacy and Language Routines</b><br><b>Think-Pair-Share</b><br>10.9.3 Collaboration provides an opportunity to leverage the Think-Pair-Share routine. In question 1, students should think through the relationship formed before discussing with a partner for question 1 part c, as well as for question 2 part b. Students should be prepared to share their thoughts during the whole-class discussion.                                                                                                                                                                                                                                | <b>Mathematical Thinking and Reasoning (MTR) Standard</b><br><b>MA.K12.MTR.1.1: Participate in Effortful Learning</b><br>In 10.9.3 Collaboration, students will work together with partners to work with angles formed by tangents and secants. During the process, students will be encouraged to stay engaged and maintain a positive mindset when working to solve tasks. |
|                                                                                                      | <br>Differentiate<br>Instruction<br><br>Consider using additional visuals, such as GeoGebra’s “An Angle Formed by Two Tangent Lines and Major and Minor Arcs” and “Angle Formed by A Secant and A Tangent,” to help students better conceptualize the concept of a tangent and its relationship to the radius in the circle for visual and kinesthetic elements. Students will have the chance to draw their own conclusions and discuss their rationale with peers and teacher, providing an opportunity for auditory learners to verbalize their conception. |                                                                                                                                                                                                                                                                                                                                                                              |
| Lesson Notes:                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                              |

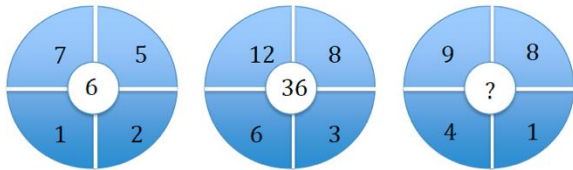
## 10.9.1 Warm-Up: Number Puzzle

**Component Overview:** Students will find the rule to determine the missing number following a number pattern.



### 10.9.1 Warm-Up: Number Puzzle

- In the circles, the outer numbers are used to calculate the number in the middle. Find the rule to determine what number should replace the question mark in the third circle.



*The number in the middle of each circle can be calculated by multiplying the difference of the two top numbers and the sum of the two bottom numbers:  $(7 - 5)(1 + 2) = 6$ ,  $(12 - 8)(6 + 3) = 36$ . The missing number is  $(9 - 8)(4 + 1) = 5$ .*



*Consider leveraging a Think-Pair-Share routine for this activity. Allow students some individual think time, and then have them work with a partner to discuss their work and rationale. Engage students in a whole-group discussion around their methods for determining the pattern.*

## 10.9.2 Exploration: Intersecting Tangent Lines

**Component Overview:** Students will explore the concept of tangent lines. For this activity, students will need a partner.

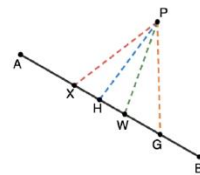


### 10.9.2 Exploration: Intersecting Tangent Lines

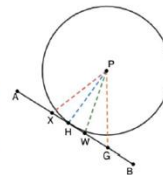
1. With your partner, answer the following.

- a. Without measuring, which dashed line represents the shortest distance from point  $P$  to line  $\overline{AB}$ ? Explain your answer.

*Answer will vary.  $\overline{HP}$  represents the shortest distance from  $P$  to line  $AB$*



- b. Point  $P$  is the center of a circle that intersects  $\overline{AB}$ . In the table, describe your observations of each dashed segment in relation to the center of Circle  $P$ .



| Segment         | Relationship with the Center of Circle $P$       |
|-----------------|--------------------------------------------------|
| $\overline{PG}$ | <i>Greatest distance to the circle center</i>    |
| $\overline{PW}$ | <i>Second-shortest distance to circle center</i> |
| $\overline{PH}$ | <i>Shortest distance to the circle center</i>    |
| $\overline{PX}$ | <i>Second-greatest distance to circle center</i> |

- c. Discuss with your partner the relationship the blue segment has with the center of Circle  $P$ .  
*The blue line looks a lot like it could be the radius of the circle.*

A **tangent** is a line or segment that intersects a circle at exactly one point.



If a line is tangent to a circle, then it is perpendicular to a radius, which ends at the point of tangency.



*What type of line represents the shortest distance between a point and a line?*



Consider using a sentence stem for students who struggle answering question 1 part c:

- If point  $H$  lies on the circle, then the blue line looks like a \_\_\_\_\_ and will be the \_\_\_\_\_ distance to  $\overline{AB}$ .

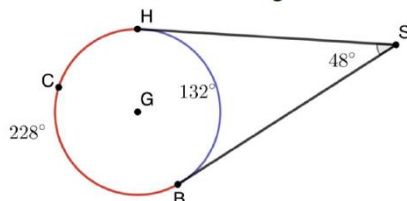
## 10.9.3 Collaboration: Angles Formed by Tangents and Secants

**Component Overview:** Students will collaborate to work with the angles formed by tangents and secants and their relationship to intercepted arcs. For this activity, students will need a partner.



### 10.9.3 Collaboration: Angles Formed by Tangents and Secants

1. Circle  $G$  with tangents  $\overline{SH}$  and  $\overline{SB}$  is shown. Use the diagram to answer the following.



- a. Complete the table.

|                      | Measure |
|----------------------|---------|
| <b>Major Arc HCB</b> | 228°    |
| <b>Minor Arc HB</b>  | 132°    |
| $\angle HSB$         | 48°     |

- b. Use the values for the major and minor arcs of Circle  $G$  to determine a relationship between the arcs and the measure of the exterior angle formed by the two tangents. The relationship can be described using words or equations.  
*The measure of the exterior angle is half the difference between the measures of the two intercepted arcs.*
- c. Ask two other partner pairs to share their relationships. Record their relationships and ask them to initial verifying you have written it correctly. If their description or equation is different than yours, discuss the differences and determine if your description or equation needs to be revised.

| Description or Equation | Initials |
|-------------------------|----------|
| Answers will vary.      |          |
|                         |          |



For a visual element, consider applets, like GeoGebra's "An Angle Formed by Two Tangent Lines and Major and Minor Arcs," for students who struggle seeing the relationship between the angle and the intercepted arc.



Consider asking the following as an opportunity to spiral back to the definitions of major and minor arcs:

- What is a minor arc? How do you name a minor arc?
- What is a major arc? How do you name a major arc?

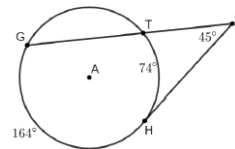
## 10.9.3 Collaboration: Angles Formed by Tangents and Secants (continued)

- d. Explain how to find the measure of an angle formed by two tangents drawn to a circle from a point outside that circle. Be sure to use the term(s) "intercepted arc(s)" at least once in the description.

*Answers will vary. The measure of an angle formed by two tangents drawn to a circle from a point outside that circle is equal to half the difference of the measures of the intercepted arcs.*

2. Circle  $A$  shows  $\angle GSH$  formed by the intersection of tangent  $\overline{SH}$  and secant  $\overline{GS}$  outside the circle at point  $S$ .  $m\widehat{GH} = 164^\circ$ ,  $m\widehat{TH} = 74^\circ$ , and  $m\angle GSH = 45^\circ$ .

- a. Use the values for  $\widehat{GH}$  and  $\widehat{TH}$  of Circle  $A$  to determine the relationship between the arcs and the measure of the exterior angle formed by the tangent and the secant.



$$\frac{164 - 74}{2} = \frac{90}{2} = 45^\circ$$

- b. Discuss the relationship with your partner.

*Students do not need to write an explanation here, but a sample response is provided for reference. It does not matter if the lines are both tangent, secant, or a combination of the two, the angle will always be equal to the difference of the two intercepting arcs divided by 2.*



The formula for the measure of an **External Angle to a Circle** formed by two tangents, two secants, or a tangent and a secant is

$$\text{Measure of angle formed} = \frac{1}{2}(\text{difference of intercepted arcs})$$

OR

$$\text{Measure of angle formed} = \frac{1}{2}(\text{larger arc} - \text{smaller arc})$$



*For a visual element, consider applets, like GeoGebra's "Angle Formed by a Secant and a Tangent," for students who struggle seeing the relationship between the angle and the intercepted arc.*

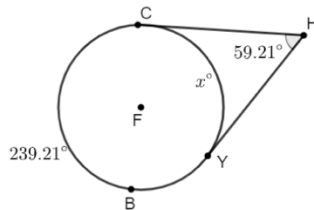
## 10.9.4 Your Turn!

**Component Overview:** Students will complete two practice problems independently on finding the missing angle measures using their knowledge of angles formed by tangent and secant lines.



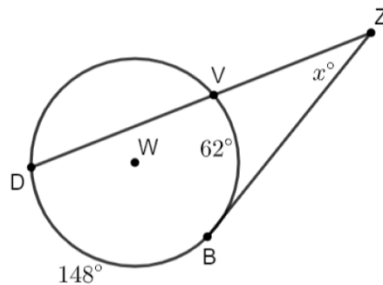
### 10.9.4 Your Turn!

1. Circle  $F$  has points  $C$  and  $Y$  on the circle. Tangents  $\overline{CH}$  and  $\overline{YH}$  intersect at point  $H$  outside the circle.  $m\widehat{CBY} = 239.21^\circ$ ,  $m\widehat{CY} = x^\circ$ , and  $m\angle CHY = 59.21^\circ$ . Find the value of  $x$ .



$$x = 120.79$$

2. Circle  $W$  has points  $D$ ,  $V$ , and  $B$  on the circle. Tangent  $\overline{BZ}$  and secant  $\overline{DZ}$  intersect at Point  $Z$  outside the circle.  $m\widehat{DB} = 148^\circ$ ,  $m\widehat{VB} = 62^\circ$ , and  $m\angle DZB = x^\circ$ . What is  $m\angle DZB$ ?



$$x = 43$$



For students who struggled in the prior lesson, use the time in this activity to work with individual students or small groups of students who may need additional support. Consider using question 1 as a “we do” style, and release students you worked with in small groups to try question 2 on their own.

## 10.9.5 Wrap-Up: Reflection

**Component Overview:** Students will complete a reflection of what they do not understand yet and what they can explain.



### 10.9.5 Wrap-Up: Reflection

Think back over this unit. If necessary, look back over the different lessons. Then, complete the sentences by checking all that apply.

*Answers will vary.*

1. I don't understand

- ☐ arc length
- ☐ area of a sector
- ☐ converting between radians and degrees
- ☐ measures of arcs and related angles

...YET.



*Consider providing students with the opportunity to expand on things they do not understand by having students write out questions they still have on Unit 10.*

2. I can explain

- ☐ how to solve problems involving arc length.
- ☐ how to solve problems involving area of a sector.
- ☐ converting angle measures between degrees and radians.
- ☐ how to solve problems involving: measures of arcs and related angles including central, inscribed, and intersections of a chords, secants, and tangents.